

MA PH 251

COMPUTER ORGANIZATION AND ARCHITECTURE

LECTURES DELIVERED BY
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First-Order Ordinary Differential Equations (FOODEs)

CHAPTER 1

Among all the mathematical disciplines, the theory of differential equations is the most important. It furnishes the explanation of all those elementary manifestations of nature which involve time

— Sophus Lie

Info:

- Next seminar (Picard Iterations and First Order ODEs)
- First midterm will contain the first three assignments

All mechanics lie on the basis of differential equations

$$\vec{F}_{\text{net}} = m \frac{d^2 \vec{r}}{dt^2}$$

All fields in physics, electromagnetism, quantum mechanics, classical mechanics, etc. are governed by differential equations which ultimately describe functions of space and time

DEFINITION 1.1 (Ordinary Differential Equation). *A differential equation involving a function of a single variable*

DEFINITION 1.2 (Partial Differential Equation). *A differential equation involving a function of more than one variable*

The order of a differential equation is determined by the greatest derivative apart of the equation. Note that higher order differential equations can be written as systems of differential equations

Reduction of order requires that the reduction occurs on the lowest order term.

Linear differential equations are differential equations for which the unknown function and its derivatives occur linearly. $\frac{d^2 x}{dt^2} = -x \frac{dx}{dt}$ is **NOT** linear, since it involves a product of a function and its derivative

The most general n^{th} order linear ordinary differential equation is

$$a_{n(x)} \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots = f(x)$$

Note, functions of x not y

A homogenous differential equation is an equation involving a term that is dependent on another variable (nonzero), inhomogenous otherwise

It is trivial that given a solution of a differential equation you can verify it is actually a solution

Finding the solution of a differential equation is a generalization in applying the Fundamental Theorem of Calculus to anti-differentiate a function

A differential equation requires that you specify for any scenario the **initial conditions** (together, these form an “Initial Value Problem”)

A partial differential equation additionally requires you to specify **boundary conditions** (specification of the values at the boundary, say for a string (is the string held in-place, free-falling, etc? (as a function of time)))

A “quasi-linear” first order ordinary differential equation is $\frac{dy}{dx} = f(x, y)$, i.e arbitrary dependence on y but $\frac{dy}{dx}$ is independent (note, how this is just specifying that $\frac{dx}{dy}$ exists)

But how do we know that any first-order ordinary quasi-linear differential equation has a solution?

When we solve differential equations, there are two problems we attempt to solve, one of **existence** and one of **uniqueness** (also one of **stability**, but we ignore them for the course)

THEOREM 1.3 (Peano). *If $f(x, y)$ is a continuous function of x and y on a closed rectangle R with sides parallel to the x - y axis and if (x_0, y_0) is any interior point on R then there exists a positive number h such that the initial value problem $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$ has a solution on the interval $[x_0 - h, x_0 + h]$*

We are not necessarily interested in this theorem since it does not guarantee uniqueness...

THEOREM 1.4 (Picard-Lindelof). *If $f(x, y)$ is continuous and $\exists L$ such that $|f(x, y_2) - f(x, y_1)| \leq L |y_1 - y_2|$ then the initial value problem has **one** solution*

This one is hard to check...

THEOREM 1.5 (Picard). *If $f(x, y)$ and $\frac{\partial f}{\partial y}$ is continuous then the initial value problem has one solution*

This one is easier to check!